Statistical Experiments 24–27 — Actual Results (Plain English)

Open the live tables canvas beside chat, or use the CSVs under `6_Results/`.

Experiment 24 — Are we over-reusing the same customers?

What we measured: For each class we take many random snapshots. Each snapshot has `t` people. We took `l = 500` snapshots. The **reuse score** `(t × l) / n■` is roughly how many times a typical person appears across snapshots. Target: about **1 or less**. Ours are much higher.

Dataset	Setting	People per class	Points per snapshot (t)	Snapshots (l)	Reuse score	OK?
Credit Card	L5 (5%)	6,630	331	500	**25.0**	No
Credit Card	L15 (15%)	6,630	994	500	**75.0**	No
German Credit	L30 (30%)	300	90	500	**150.0**	No
German Credit	L60 (60%)	180	180	500	**300.0**	No

Suggested snapshots if we keep `t` the same:

Dataset	Setting	Current l	Suggested l
Credit Card	L5	500	**20**
Credit Card	L15	500	**7**
German Credit	L30	500	**3**
German Credit	L60	500	**2**

Files: `6_Results/24_Sampling_Ratio_Audit/sampling_ratio_audit.csv`, `suggested_l_values.csv`

Experiment 25 — Average barcode feature values across snapshots

What we measured: Each snapshot → 24 numbers. Across 1,000 snapshots we stored mean and variance of every feature.

Highlights:

Dataset / setting	Feature	Mean	Variance
Credit Card L5	Mean Death (H■)	0.193	0.00013
Credit Card L5	Mean Persistence (H■)	0.023	0.000003
Credit Card L15	Mean Death (H■)	0.144	0.00007
Credit Card L15	Mean Persistence (H■)	0.018	0.000001
German Credit L30	Mean Death (H■)	1.121	0.00135
German Credit L30	Mean Persistence (H■)	0.114	0.00015
German Credit L60	Mean Death (H■)	0.996	0.00035
German Credit L60	Mean Persistence (H■)	0.110	0.00005

Full 96-row table: `6_Results/25_Snapshot_Mean_Variance/snapshot_mean_variance.csv`

Experiment 26 — How “high-dimensional” is the data?

What we measured: Intrinsic dimension with Two-NN (main number). Roughly: how many degrees of freedom the cloud needs. Concern from the team: if this is ~7 we may be in trouble. It is **not**.

Dataset	Two-NN (all features)	Two-NN (after PCA)	PCA components used in TDA	Variance kept
Credit Card	3.95	**2.81**	7	94.0%
German Credit	5.34	**4.06**	15	89.3%

File: `6_Results/26_Intrinsic_Dimension_Estimation/intrinsic_dimension_estimates.csv`

Experiment 27 — Do default and non-default snapshots differ?

What we measured: Permutation test comparing default vs non-default snapshot clouds.

p = 0.005 means: under 200 random label shuffles, the real split looked more separated than almost all shuffles → classes look different.

Dataset	Test	Observed score	Mean if labels shuffled	p-value	Verdict
Credit Card L5	F■,■	0.0109	0.0131	0.005	Differ
Credit Card L5	F■,■	0.2200	0.2888	0.005	Differ
Credit Card L15	F■,■	0.0069	0.0089	0.005	Differ
Credit Card L15	F■,■	0.1611	0.2364	0.005	Differ
German Credit L30	F■,■	0.1015	0.1139	0.005	Differ
German Credit L30	F■,■	1.0151	1.0662	0.005	Differ
German Credit L60	F■,■	0.0454	0.0560	0.005	Differ
German Credit L60	F■,■	0.6310	0.6995	0.005	Differ

File: `6_Results/27_Null_Hypothesis_Algorithm2/algorithm2_permutation_results.csv`

Bottom line

1. **500 snapshots is too many** — reuse scores are 25×–300×; suggested `l` is 2–20.
2. **Barcode features are stable** — means logged; variances small.
3. **Intrinsic dimension (Two-NN) is below 7** on both datasets.
4. **Default vs non-default snapshots differ** on every test we ran (p = 0.005).